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Unit - 2

↳ Read Software Requirement Specification shared in Google Classroom

31/08/26

System model also known as context level diagram.

Semantic versioning → different versions are created only if it is meaningful.

build/release/version difference

02/09/26

UML (Unified Modelling Language)

JML 5.x

* It is a standard visual language used to model, visualise & document a S/W system.
* Instead of explaining a software system only through text or code, UML allows us to represent it visually
* For example, if a Functional Requirement (FR) says a student can register for a campus event, this is understandable in text but it doesn't tell us the following -

(1) Who interacts with the system?
(2) What exactly happens?
(3) What information is involved?

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(4) Which objects are communicated?

(5)

* UML is a way of thinking about the S/W system before or alongside the implementation.

* UML Diagram Selection

Model	Question
(1) Scenario	What happens in a particular situation?
(2) Use case	Who uses the system & what do they want?
(3) Activity	What is the workflow?
(4) Class	What concepts or classes exist?
(5) Object	What actual instances exist?
(6) Collaboration	Which objects communicate?
(7) State	How does an Object change?
(8) Component	What S/W units make up the system?

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* UML has 2 categories —

Structural
↳ addresses the question what exist?
↳ class, object & component

Behavioural
↳ addresses what happens?
&
↳ scenario, activity, state & collaboration

① Scenario

* It is a specific sequence of events & interactions that describes one particular situation in the system.

* For example, if FR says Register for event, this is a broader functionality for "A student wants to attend a technical workshop and register for it to Campus Hub".

* Requirement: It is a general state of need.

* Scenario: It is a specific sequence of interaction.

For eg: Requirement: Student should be able to register for an event.

Scenario: Student selects an available event, clicks register, the system validates the request, creates the registration & displays confirmation.

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Scenario: Successful event registration.
Actor: student
Goal: successful registration of event
Precondition: * student is logged in.
* ~~student~~ event exists.
* registration is open
Main Flow: 1. Student logs into CampusHub
2. CampusHub displays the available events
3. Student selects an event.
4. CampusHub provides/displays the event details.
5. Student clicks on register.
6. CampusHub checks if registration is open.
7. CampusHub checks if student is eligible.
8. CampusHub checks whether seats are available,
9. CampusHub creates the registration.
10. CampusHub displays the registration confirmation.

Representation:

vertically only

Student
↓
login
↓
Browse events
↓
select event
↓
view details
↓
Register

check eligibility
check seat availability → create registration → registration conformation

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12 group UML State UML Diagram: 10

Alternative scenarios: ① Event is full. → ⑤ Alternative flow

② Registration is closed.

③ Student is

④ Student receives registration confirmation.